Chapter 5
Computer contracts

Computing has presented far fewer problems for the law of contract than it has in other areas of the law such as intellectual property, discussed in Chapter 6, and in the criminal law, which forms the subject matter of Chapter 11. There are a number of reasons for this. Contracts set out the agreement between the parties: they set out the aims of the parties; provide for matters arising while the contract is running, ways of terminating the contract and the consequences of termination. Where there are gaps in the agreement because the parties have failed to contemplate a particular issue, it is a function of contract law to fill them, for example by implying terms; also contract law provides rules for the termination of the contract if performance becomes impossible; and sometimes, although fairly rarely, it sets aside contracts which are too harsh or unconscionable.

There are almost never disputes over contracts which run perfectly. An analogy can be made with a good marriage, where there is no need for the law to intervene. But, if things go wrong, in a contract or in a marriage, the law provides a framework for the settlement of points of disagreement, and for the termination of the relationship.

Contract law since its inception has handled disputes. An example can be given of a ship chartered to carry a cargo, where, for instance, the cargo rots before reaching port, or the ship sinks, or the ship and its cargo are impounded in the course of a war. What are the rights and remedies of the parties to the charter agreement? Is there a contract of insurance covering the loss of the goods or the ship, and is liability for any matter excluded under the insurance contract, such as loss caused through warfare? Agreements for the provision of goods and services etc. connected with computing present no insuperable problems for a set of laws which has been regulating commercial dealings and handling disputes for many hundreds of years.

One noticeable growth area is e-commerce. However, again the law is able to cope, sometimes with minor modifications, e.g. relating to electronic signatures to documents exchanged over the Net. Since the advent of the Internet, the market has globalized to a far greater extent than ever before, and there is greater need for international harmonization of laws. The EU has therefore been very active in line with its policy of removing distortions of trade within the internal market and also in facilitating trade by EU businesses. There are therefore directives and proposals for directives on:

1. legal protection for encrypted services in the internal market;
2. electronic signatures;
3. electronic commerce;
4. distance contracts;
5. distance selling of financial services.

One of the problems with computing contracts is that many lawyers are still not familiar

with the technology. But, on the other hand, even fewer computer scientists are familiar with the law; and as both lawyers and computer scientists use jargon known almost only to themselves, the difficulties are compounded. These difficulties are however receding, for more lawyers are becoming familiar with computing through use of computers at home and in work; there are more lawyers specializing in this area of the law; and many books have been written describing the framework of computer contracts, and providing model contracts or precedents, which lawyers can adapt to the needs of their clients. Let us hope that more computer scientists will become familiar with the law.

Hilary Pearson [1] made a very telling statement when she said that, while optimists make the best deal makers, pessimists make the best contract writers. When it comes to drafting a contract, lawyers in particular are born pessimists. This often gives rise to frustration on the part of a business client who is excited by the possibilities of a deal which may have taken considerable time and effort to negotiate. Resolving potential and hypothetical points of difficulty may just be seen as time-wasting by the client.

It is possible to avoid formal written contracts for the supply of goods or services, and to work on a basis of oral discussions only, or on the basis of a set of loosely drafted heads of agreement, but for the security of all parties to the deal, it is much better to set out clearly the terms on which each is to work, and to provide for such matters as the method of payment; what happens if there is a defect in the goods, software or services provided. It is also advisable to settle the appropriate methods of terminating the agreement—for instance, the length of notice required to terminate the agreement; and whether the length of notice should vary at different times in the course of the agreement, or on the occurrence of certain events. Drafting a document which sets out all of these matters takes time, but in many cases it is better to pause and think it

1. Pearson, *Computer contracts* (London: Financial Training), 1984.

out in advance, than it is to face difficulties later, when commercial and other factors may change and when all may not appear to be as clear-cut or financially beneficial as was originally thought.

It is important that a contract is set out in a clear and logical manner and that it is complete and consistent. There should be no ambiguity and the parties to the agreement should be left in no doubt as to their rights and duties. Ambiguity and doubts can lead to performance which is viewed as unsatisfactory. This can lead to disagreement and the expenditure of time, effort and therefore money, in resolving the matter. This is the last thing that the parties to a commercial agreement need and want. As stated earlier, it is much better to avoid the problems initially by careful preparation and drafting. However, if there is a dispute, and often these are unavoidable because they are caused by circumstances beyond the control of the contracting parties, it is again important that the contract is properly drawn up, for at this stage it will be interpreted by outsiders, for example by lawyers and judges, or maybe by accountants or a trustee in bankruptcy if one of the parties has become insolvent; and outsiders, even more than the contracting parties themselves, will need to be able to envisage and understand the intricacies of the deal, and will require clarity and consistency in the documentation.

In the course of their work, software engineers are likely to come across many different

types of contract—insurance contracts, contracts of employment, contracts with hardware suppliers, consultancy contracts and so on. In this chapter we shall concentrate on contracts which are relevant to the development and supply of software; in particular, we shall describe in some detail the elements which go to make up contracts for the supply of software which is to be custom-built for a client for a fixed price, for this illustrates most of the issues which are likely to arise in contracts regarding the provision of software and related services. Other specialist works should be consulted by those interested in the other types of contracts. A number of these specialist works are listed in the section on Further Reading at the end of this chapter.

5.1 Contracts for the supply of custom-built software at a fixed price

5.1.1 Structure of the contract

Producing a good contract costs a lot of money; good commercial lawyers are not cheap. For this reason, software suppliers try to use what are known as standard form contracts, which are used or intended to be used many times over. Such a contract might consist of:

- a short introductory section, which specifies, among other things, the names of the parties to the contract;
- a set of standard terms and conditions;
- a set of appendices or annexes.

The standard terms and conditions do not change from one project to another; they contain references to the annexes, which contain all the project specific material.

5.1.2 The introductory section

The first part of the contract is brief; it states that it is an agreement between the parties whose names and registered addresses are given. It is dated and signed by authorized representatives of the parties. It often begins with a set of definitions of terms used in the course of the agreement, set out either in alphabetical order, like a dictionary, or in the order in which they appear in the rest of the contract. These definitions explain precisely what the parties mean by certain words or phrases. Once the term is defined, its use elsewhere in the contract should conform only to that definition, thereby ensuring consistency and avoiding ambiguity. Definitions are also useful in cutting down descriptions elsewhere and avoiding the need to change the standard terms and conditions. For example, the definitions section will tell us that Company X Ltd, the software house, is to be referred to throughout the contract as “The Company”, and Company Y Ltd, which has commissioned the work, is to be known throughout as “The Client”.

It is also important that the introductory section states that the contract consists of the introductory section itself, the standard terms and conditions, the annexes (which should be listed), together with any documents listed in the annexes, such as the requirements specification (discussed in the paragraph below) and nothing else. This is to avoid, for

example, the situation in which statements made by an over-enthusiastic salesman while trying to win the business are claimed by the client to constitute part of the contract.

The following subsections discuss the issues which must be addressed by the standard terms and conditions.

5.1.3 What is to be produced

It is clearly necessary that the contract states what is to be produced. There are usually two levels of reference here: the standard terms and conditions refer to an annex and the annex then refers to a separate document which constitutes the requirements specification. It is important that the reference to the requirements specification identifies that document uniquely; normally this will mean quoting a date and issue number.

Software engineers will be familiar with the problems of producing requirements specifications. A specification sets out the detailed requirements of the client. Ideally, the specification should be complete, consistent and accurate and set out all that the client wants to be done in the performance of the contract. Unfortunately, we know that it is very difficult to achieve this ideal standard and, even if we succeed, the requirements of the client may evolve as the contract proceeds, and sometimes the changes may be substantial. How are these changes to be accommodated by a contract which, in a sense, freezes the requirements of the parties to those at one particular time by incorporating the original specification into the contract? The answer is that the contract should provide a procedure for making variations to the specification or job description, then follow this through by providing a method of calculating payment for work done to facilitate the changes, and also perhaps provide for a variation of the level of anticipated performance, and maybe also vary the method of acceptance testing. In other words, once again, the contract should anticipate events and provide an agreed formula for modification.

5.1.4 What is to be delivered

Producing software for a client is not, usually, a matter of simply handing over the text of a program which does what is required. It is important, therefore, that the contract states (usually in an annex) what precisely is to be provided. The following is a non-exhaustive list of possibilities:

- source code;
- command files for building the executable code from the source and for installing it;
- documentation of the design and of the code;
- reference manuals, training manuals and operations manuals;
- software tools to help maintain the code;
- user training;
- training for the client's maintenance staff;
- test data and test results.

5.1.5 Ownership of rights

It is important that the contract should also state just what legal rights are being passed by

the software house to the client under the contract. Ownership in physical items such as books, documents or discs will usually pass from the software house to the client, but other intangible rights, known as intellectual property rights, present more problems. Intellectual property law forms the subject matter of Chapter 6. As we shall see there, software is potentially protectable by a number of intellectual property rights, such as copyright, design rights, confidentiality and trade marks. It is important for the contract to state precisely who is to own these rights. Do they pass to the client or are they retained by the software house?

Let us take copyright as an example. Copyright is owned by the author of a work, which in the case of a literary work, or a computer program which is not computer generated, is the person who creates it. All of these technical terms are explained in Chapter 6 to which reference should be made. But if the person who creates the program is an employee acting in the course of employment, then under Section 11 of the Copyright, Patents and Designs Act 1988, the copyright belongs to the employer. Therefore, if a software house employs a programmer, copyright in the programs belongs to the software house. On the other hand, if a program is written for a software house by an outside consultant or independent contractor, who is not one of their employees, Section 11 will not apply, and, as is explained in Chapter 6, the consultant owns the copyright. Ownership in copyright passes only by written assignment or transfer. If the consultant owns the copyright, the software house should consider whether it wants ownership of the copyright either in order to retain it itself, or in order to pass it on to the client. If ownership of the copyright is to pass from the consultant to the software house, a term to that effect should be inserted into a written contract between the consultant and the software house. An oral agreement to transfer the property right is insufficient.

The next decision to be taken is whether the copyright in the program should pass from the software house to the commissioning client; or whether it should remain with the software house. If ownership of copyright passes to the client it is known as a sale or assignment and again a written agreement is necessary. Furthermore, the agreement will usually provide that copyright is only to pass to the client when the final payment has been made in full. If copyright is to remain with the software house and the client is merely given permission to use the software, this is known as a licence.

If a commercial firm commissions a software house to write a program especially for it, the client will often want exclusivity. This can be achieved in a variety of ways. Normally the client will want to be the outright owner of the copyright, so that others can be prevented from using it. Another way to achieve exclusivity is to ask for an exclusive licence to use the copyright, ownership of which remains in the software house. If the client has an exclusive licence to use the software, it is the only organization entitled to use it. If the client takes ownership of the software or has an exclusive right to use it, the software house cannot make money from the software by licensing others to use it. The software house must therefore make its entire profit on the work from that particular deal. The cost to the client is therefore at its highest. If, however, the software house retains the right to allow others to use it, the costs are spread more widely and the cost to the commissioning client may be diminished. In these cases, ownership in the copyright in the software remains with the software house and the commissioning client is given a non-exclusive licence. The commissioning client may still retain some control over the

extent to which the software is distributed to others by retaining a right to veto the grant of licences to others. Care must, however, be taken as the veto could be used to stifle competition and may fall foul of competition laws.

Where the client is granted a licence, the following matters should be dealt with in the contract:

1. duration of the licence—a licence should be for a fixed period; or there should be some provision for termination, for example by giving notice, or on the happening of certain events, common terminating events being death, or insolvency;
2. the licence agreement should state whether the licensee can assign or transfer the licence to another. If there is no provision giving the licensee the power to transfer the licence to another, then the licence is probably not assignable;
3. scope of the licence: does the licence cover use on one particular computer, or can the software be run on other machines. If so, is the licence limited to one site? If the client is one of a group of companies, can others in the group also benefit from the licence?
4. confidentiality: the licence will often seek to restrain the licensee from allowing anyone other than company employees to become familiar with the use of the software. This can be an embarrassment for educational establishments who wish to purchase the software for use by their students.

If the supplier retains the copyright, major problems can arise for the client if the supplier goes into liquidation or otherwise ceases to trade. The supplier is then no longer able to maintain the software but the client may be unable to obtain copies of the up-to-date source listings of the programs and any tools used to construct them, in order to commission maintenance from a third party. One way around this difficulty is for the contract to specify that, after acceptance, a copy of the listings and documentation is placed in escrow; this means that the copy is placed in the hands of a third party (usually a lawyer) to be released to the client if and when certain defined circumstances arise. The escrow copy is replaced whenever the software is updated.

5.1.6 Confidentiality

A second area of intellectual property law which should be considered in a software contract is confidentiality. The commissioning client may well have to pass confidential information about its business operations to the software house. On the other side of the coin, the software house may not want the client to divulge to others details of the program content or other information gleaned about its operations by the client. It is usual in these circumstances for each party to promise to maintain the confidentiality of the other's secrets, and for express terms to that effect to be included in the contract. Confidentiality is discussed in the chapter on intellectual property rights (Chapter 6), to which reference should be made.

5.1.7 Payment terms

The standard terms and conditions will specify the payment conditions, that is something along the lines that:

> Payment shall become due within thirty days of the date of issue of an invoice. If payment is delayed by more than thirty days from the due date, the Company shall have the right, at its discretion, to terminate the contract, or to apply a surcharge at an interest rate of 2 per cent above the bank base lending rate.

In practice, such clauses are only brought into effect in extreme cases, since using them is likely to destroy the goodwill between supplier and client on which the success of the project depends.

It would be unusual, in a project of any significant length, for all payment to be delayed until the work is complete and accepted. An annex will usually specify a pattern of payments like the following:

1. an initial payment of, say, 15 per cent of the contract value becomes due on signature of the contract;
2. further stage payments become due at various points during the development, bringing the total up to, say, 65 per cent;
3. a further 25 per cent becomes due on acceptance of the software;
4. the final 10 per cent becomes due at the end of the warranty period.

Such a pattern has advantages for the supplier in that it reduces the financial risk arising from possible insolvency of the client or from default for other reasons and it reduces possible cash flow difficulties. If the client is not prepared to accept a payment pattern of this type, the supplier is likely to demand a premium to cover the increased risk and the costs of financing the development. In negotiating the payment pattern, the supplier will usually seek to have the stage payments becoming due on fixed calendar dates while the client will try to have them tied to the achievement of specific project milestones, e.g. approval of the design specification.

5.1.8 Calculating payments for delays and changes

It happens not infrequently that progress on the development of a piece of software is delayed by the failure of the client to meet obligations on time. While the supplier will be expected to use its best endeavours to rearrange activities so as to avoid wasting effort, this is not always possible. The contract should therefore make provision for payments to compensate for the wasted effort, incurred, for example, when the client fails to provide information on a due date or when changes are requested which result in extra work.

The contract must specify the process by which these extra payments are to be calculated. Typically, an annex will include daily charging rates for each grade of staff employed on the contract and the amount of extra effort to be paid for will be agreed at progress meetings.

Delay payments and payments for variations to the original requirements are, perhaps, the commonest cause of contractual disputes, not only in software engineering but in most other contracting industries—the construction industry is a notorious example. One reason for this is that competitive bidding for fixed price contracts often means that the profit margin on the original contract is very low so that companies seek to make their profit on these additional payments.

5.1.9 Penalty clauses

The previous subsection dealt with compensation for delays caused by the client; delays caused by the supplier are handled differently.

The normal mechanism used is to include a penalty clause which provides that the sum payable to the supplier is reduced by a specified amount for each week that acceptance of the product is delayed, up to a certain maximum. Thus, on a contract of value £1 million, the penalty might be specified as £5,000 per week up to a maximum of £100,000.

Delays in delivering working software are notoriously common; it might therefore be expected that contracts for the supply of software would normally include such a penalty clause. Paradoxically, such provision is comparatively rare. There are three reasons for this:

- Suppliers are very reluctant to accept penalty clauses and anything stronger than the example quoted above is likely to lead to reputable suppliers refusing to bid.
- If the contract is to include penalty clauses, the bid price is likely to be increased by at least half the maximum value of the penalty.
- If the software is seriously late and penalties approach their maximum, there is little incentive for the supplier to complete the work since he will already have received in stage payments as much as he is going to get.

It should be realized that the cost of delays on fixed price contracts is very high, regardless of penalty payments. Every delay eats into the supplier's profit margin. As a result, suppliers are strongly motivated to produce the software on time and delay is usually the result of genuine technical difficulties (or incompetence!) rather than lack of motivation.

5.1.10 Obligations of the client

In almost all cases where work is being carried out for a specific client, the client will have to fulfil certain obligations if the contract is to be completed successfully. The following is a (non-exhaustive) list of possibilities:

- provide documentation on aspects of the client's activities or the environment in which the system will run;
- provide access to appropriate members of staff;
- provide machine facilities for development and testing;
- provide accommodation, telephone and secretarial facilities for the company's staff when working on the client's premises;
- provide data communications facilities to the site.

The general terms and conditions will normally state that a list of specific obligations and the dates at which they will be required is given in an annex. It will also state that failure to meet these obligations may render the client liable for delay payments.

5.1.11 Standards and methods of working

The supplier is likely to have company standards, methods of working, quality assurance procedures, etc. and will normally prefer to use these. More sophisticated clients will have their own procedures and may require that these be adhered to. In some cases, the supplier may be required to allow the client to apply quality control procedures to the project. The contract must specify which is to apply.

5.1.12 Progress meetings

Regular progress meetings are essential to the successful completion of a fixed price contract and it is advisable that standard terms and conditions require them to be held. The minutes of progress meetings, duly approved and signed, should have contractual significance in that they constitute evidence that milestones have been reached (so that stage payments become due) and that delay payments have been agreed.

5.1.13 Project Managers

Each party needs to know who, of the other party's staff, has day-to-day responsibility for the work and what the limits of that person's authority are. The standard terms and conditions should therefore require each party to nominate, in writing, a Project Manager. The Project Managers must have at least the authority necessary to fulfil the obligations which the contract places on them. It is particularly important that the limits of their financial authority are explicitly stated, i.e. the extent to which they can authorize changes to the cost of the contract.

5.1.14 Acceptance procedure

Acceptance procedures are a critical part of any fixed price contract for they provide the criteria by which successful completion of the contract is judged. The essence of the acceptance procedure is that the client should provide a fixed set of acceptance tests and expected results and that successful performance of these tests shall constitute acceptance of the system. The tests must be provided at or before the start of the acceptance procedure; within reason, there may be as many tests as the client wishes but extra tests cannot be added once the test set has been handed over. The purpose of this restriction is to ensure that the acceptance procedure can be completed in reasonable time.

Other points to be addressed under this heading include who shall be present when the tests are carried out and what happens if the tests are not completed successfully.

5.1.15 Warranty and maintenance

Once the product has been accepted, it is common practice to offer a warranty period of, typically, 90 days. Any errors found in the software and reported within this period will be corrected free of charge. This clause is, of course, subject to negotiation; reducing or eliminating the warranty period will reduce the overall cost of the contract and

prolonging the period will increase it.

Once the warranty period is over, the supplier may offer, or the client demand, that maintenance will continue to be available on request. Since such maintenance is likely to involve enhancement of the software rather than simply correction of faults, the resources required are unpredictable—the client almost certainly does not know what enhancements will be required in two years' time. For this reason, a fixed price for the maintenance will not be appropriate. Maintenance will therefore usually be charged on a time and materials basis; the client may possibly be required to commit to taking a fixed number of days of effort each year in order to compensate the supplier for the need to retain knowledge of the system.

5.1.16 Indemnity

It could happen that, as a result of the client's instructions, the supplier is led unwittingly to infringe the intellectual property rights of a third party or that, through carelessness or dishonesty, the supplier provides a system which infringes such rights—perhaps through using proprietary software as a component of the system delivered. For this reason, it is advisable to include a clause under which each party indemnifies the other for liability arising from its own faults in this respect.

5.1.17 Termination of the contract

There are many reasons why it may become necessary to terminate a contract before it has been completed. It is not uncommon, for example, for the client to be taken over by another company which already has a system of the type being developed, or for a change in policy on the part of the client to mean that the system is no longer relevant to its needs. It is essential, therefore, that the contract make provision for terminating the work in an amicable manner. This usually means that the supplier is to be paid for all the work carried out up to the point where the contract is terminated, together with some compensation for the time needed to redeploy staff on other revenue-earning work. The question of ownership of the work so far carried out must also be addressed.

5.1.18 Arbitration

Court action to resolve a contractual dispute is likely to be expensive. For this reason, it is common practice for contracts to include a statement that, in the event of a dispute that cannot be resolved by the parties themselves, they agree to accept the decision of an independent arbitrator. Provision is usually made for the arbitrator to be appointed either by the President of the British Computer Society or by the President of the Institution of Electrical Engineers. Both bodies maintain lists of qualified arbitrators who have the necessary technical understanding.

An arbitration clause will usually state that, if arbitration is required, it will take place in accordance with the Arbitration Act 1996. This Act of Parliament lays down a set of rules for arbitration that cover many eventualities and reference to it avoids the need to spell these out in detail; most of the provisions of the Act are optional, in the sense that

they come into effect only if the contract contains no alternative provision.

Some organizations may be unwilling to accept an arbitration clause because they feel that they are signing away some of their legal rights.

5.1.19 Inflation

In lengthy projects or projects where there is a commitment to long term maintenance, the supplier will wish to ensure protection against the effects of unpredictable inflation. To handle this problem, it is customary to include a clause which allows charges to be increased in accordance with the rise in costs. The Government publishes several different financial indices; the Retail Price Index is the most widely known but the most appropriate one for the purposes of this type of contract is the Business Costs Index.

The clause should state how often (once a year, twice a year) charges can be increased and how the effect on the overall price is to be calculated.

5.1.20 Applicable law

Where the supplier and the client have their registered offices in different legal jurisdictions or performance of the contract involves more than one jurisdiction, it is necessary to state under which laws the contract is to be interpreted.

5.2 Other types of software services contract

There are four types of contractual arrangement which are widely used in connection with the provision of software services:

- contract hire;
- time and materials;
- consultancy;
- fixed price, as described in the previous section.

5.2.1 Contract hire

Under a contract hire agreement, the supplier agrees to provide the services of one or more staff to work for the client; the staff work under the direction of the client and the supplier's responsibility is limited to providing suitably competent people and replacing them if they become unavailable or are adjudged unsuitable by the client. Payment is on the basis of a fixed rate for each man day worked; the rate depends on the experience and qualifications of the staff. Contract hire is sometimes referred to disparagingly as "body shopping". Closely related are the freelance agreements under which individuals sell their own services to clients on a basis similar to contract hire.

Contract hire agreements are very much simpler than fixed price contracts because the supplier's involvement and responsibility are so much less. Issues such as delay payments, acceptance tests and many others simply do not arise; however, as mentioned earlier, ownership of intellectual property rights generated in the course of the work must

be addressed.

5.2.2 Time and materials

A time and materials contract (often referred to as a "cost plus" contract) is somewhere between a contract hire agreement and a fixed price contract. The supplier agrees to undertake the development of the software in much the same way as in a fixed price contract but payment is made on the basis of the costs incurred, with labour charged in the same way as for contract hire. The supplier is not committed to completing the work for a fixed price, although a maximum payment may be fixed beyond which the project may be reviewed. Many of the complications of fixed price contracts still occur with time and materials contracts—ownership of rights, facilities to be provided by the client, progress monitoring arrangements, for instance—but others, such as delay payments and acceptance testing do not; this is not to say that no acceptance testing is done, only that it has no contractual significance since nothing contractual depends on its outcome.

It may be wondered why any client should prefer a time and materials contract to a fixed price contract—surely it is better to have a contract which guarantees performance for a fixed price rather than one in which the price is indeterminate and there is no guarantee of completion? In the first place, it often happens that the work to be carried out is not sufficiently well specified for any supplier to be prepared to offer a fixed price; part of the supplier's task will be to discover what is required and to specify it in detail. Secondly, a supplier always loads a fixed price contract with a contingency allowance, to allow for the risk that unexpected factors will cause the project to require more resources than originally estimated. If all goes well, the supplier makes an extra profit; this is the reward for risk taken. By accepting a time and materials contract, this risk and the possibility of extra profit (in the form of a lower cost) are effectively transferred to the client, who also avoids the dangers of having to pay excessive sums to have minor changes incorporated into the specification. All this having been said, it remains the case that there is a strong movement away from time and materials towards fixed price, noticeably in the defence field.

5.2.3 Consultancy contracts

The use of consultants is now widespread both in private industry and in public bodies. Consultants are typically used to assess some aspect of an organization and to make proposals for improvements. The end product of a consultancy project is therefore usually a report or other document. Consultancy projects are usually undertaken for a fixed price but the form of contract is very much simpler than the fixed price contracts so far described.

There are two reasons for the comparative simplicity of consultancy contracts. First, the sums of money are comparatively small and neither side stands to lose a great deal. Second, while it is possible to demonstrate beyond doubt that a piece of software does not work correctly and, thus, that the supplier has failed to fulfil the contract, it is not usually possible to demonstrate unequivocally that a report fails to fulfil a contract. The client has to rely on the desire of the supplier to maintain a professional reputation and in practice

this usually proves sufficient to ensure that the work is of an acceptable standard.

It should be obvious that confidentiality is of particular importance in consultancy contracts. It is also important that the contract refers explicitly to the terms of reference of the consultancy team and, in practice, these are perhaps the commonest source of disagreements in consultancy projects. As a result of their initial investigations, the consultants may discover that they need to consider matters which were outside their original terms of reference but the client may be unwilling to let this happen for any one of a number of possible reasons.

5.3 Liability for defective software

The law in this area is complex. It is dealt with only briefly here. A fuller discussion is to be found in Reed (ed.): *Computer Law* (4th edn.); and in Rowland and Macdonald: *Information Technology Law,* both of which are noted in the Further Reading section at the end of this chapter; and you should refer also to Chapter 10 of this book.

Firstly we should look at terms implied into a contract by the law. Just which terms are implied will depend on the type of contract under consideration. If it is a contract for the sale of goods, the Sale and Supply of Goods Act 1994 implies that the goods shall be of satisfactory quality and fit for their purpose. Section 1(2) of the 1994 Act, which amends Section 14 of the Sale of Goods Act 1979, provides that goods are of satisfactory quality if they meet the standard that a reasonable person would regard as satisfactory, taking into account any description of the goods, the price (if relevant) and all other relevant circumstances. The quality of goods includes their state and condition and the following (among others) are in appropriate cases aspects of the quality of the goods:

(a) fitness for all the purposes for which goods of the kind in question are commonly supplied;
(b) appearance and finish;
(c) freedom from minor defects;
(d) safety;
(e) durability.

There is no breach of the implied term if a matter which might make the goods unsatisfactory is specifically drawn to the buyer's attention before the contract is made.

A major difficulty is whether software constitutes goods for the purposes of the Sale of Goods Act 1979 and the 1994 Act. Goods are defined in Section 61 of the 1979 Act in a way which covers physical (tangible) items such as books, discs and hardware but may well exclude intangible items such as software [2] . However, it is arguable that software on a disc, particularly mass-produced software, where the value of the software in itself is not too great per individual disc, might come within the definition of "goods" in much the same way as does music on a cassette, the images on a video tape, or the words in a book. It has long been recognized that a book, which has some unprinted pages or contains pages from another book instead of the one requested, is a defective item of goods. By analogy, a defective piece of software might also be caught by the legislation. But while this argument might apply to mass-produced software such as computer games and

common word-processing packages the position of bespoke software is probably quite different.

A contract for the supply of bespoke software is more appropriately classified as a contract for work and materials. The work, which has gone into producing the software, in the form of research and development, is the most important component of the contract. Goods (or materials) supplied, such as tapes, discs or manuals are subsidiary items. The distinction between a contract for the sale of goods and one for work and materials is essentially one of emphasis. The classification of customized software might well be quite difficult. However, the legal consequences of the classification are important, for if the contract is one for work and materials, the Sale of Goods Act will not apply. Instead the contract falls under the Supply of Goods and Services Act 1982, and this Act implies into the contract a term that the software must have been written with reasonable skill and care. This is a notoriously vague notion. Only a moment's thought will show that it does not accord a great deal of protection for the commissioning client. It in no way guarantees that the software is "bug-free"; only perhaps that the number of bugs, or the problems caused by the bugs, is no more than could reasonably be expected from the reasonable software house producing bespoke software to those specifications at that time. Because of the difficulty in applying this standard, a contract will often contain an express guarantee, for example to cure defects for a certain length of time after delivery. Guarantee clauses should be carefully drafted for the software house and carefully construed by the commissioning client, for obviously much will depend on the precise wording used.

The foregoing account has considered the liability of a software house in contract law. There are of course other ways in which a software house can be rendered liable for defective software which causes loss or damage [3] , for example, the tort of negligence, or the law relating to negligent misstatement might

2. See also Chapter 10.
3. For damage which causes physical injury see Chapter 10.

apply. These might cover either physical damage or financial loss, although it is generally more difficult to recover for financial loss in negligence than it is to recover for physical damage to the person or to property. Also, it is difficult to prove negligence. For this reason, and to conform to an EC Directive, a strict liability regime for property damage and personal injury was introduced by the Consumer Protection Act 1987 under which it is not necessary to prove fault in order to establish liability. There are, however, difficulties in applying this Act [4] . These matters are outside the scope of this chapter, the aim of which is to introduce computer contracts.

5.3.1 Exclusion of liability

The aim of a party drafting a contract will be to minimize liability under the contract, or to exclude it altogether. The law is governed by the Unfair Contract Terms Act 1977. Briefly, the 1997 Act provides that liability for death or personal injury can never be excluded. Liability for other forms of damage (such as property damage) can be

excluded, but only in so far as it is reasonable to do so. Liability under the terms implied under the Sale of Goods Act 1979 (merchantability or fitness for purpose) or under the Supply of Goods and Services Act 1982 (to use reasonable skill and care) to which we referred earlier cannot be excluded in a consumer sale, but in a non-consumer sale they can be excluded if the exclusion is reasonable. In order for the sale to be a consumer sale, the requirements of Section 12 must be satisfied. This means that the buyer must be a private person; the buyer must buy from a seller who is acting in the course of a business; and the goods must be of a type ordinarily supplied for private use or consumption. If I buy a computer game from W.H.Smith to play at home on my microcomputer, Section 12 is satisfied, and under the 1977 Act the implied terms cannot be excluded. However, where the client commissions bespoke software from a software house in order to use it in the course of business, this is not a consumer sale, and the implied terms can be excluded if it is reasonable to do so. Reasonableness is ascertained by reference to guidelines laid down in Schedule 2 to the Unfair Contract Terms Act. Matters to which the court would have regard in assessing whether it is reasonable for a supplier to exclude liability include:

- the relative bargaining powers of the parties;
- opportunities to contract with others who do not use the same exclusionary terms;
- whether the customer knew or ought reasonably to have known of the existence and extent of the term (note here that in practice contracting parties should always read all contracts before entering into them and

4. See further Chapter 10.

should seek advice on matters such as the extent of any exclusion clause or clause seeking to limit liability);
- whether the goods were manufactured, processed or adapted to the special order of the customer.

5.3.2 Standard form contracts and limitation of liability

It was stated at the start of this chapter that a software house will often try to use a standard form contract in its dealings with its commissioning clients. Section 3 of the Unfair Contract Terms Act contains special provisions for standard form contracts. In the present context it provides that a software house using a standard form contract cannot, again unless it is reasonable to do so,

(i) exclude or restrict liability for its own breaches of contract; or
(ii) claim to be entitled

(a) to render a contractual performance substantially different from that which was reasonably expected of it; or
(b) in respect of the whole or any part of its contractual obligation, to render no performance at all.

If a standard form contract limits liability in damages to a fixed or determinable sum,

Section 11(4) applies. This provides that:

> Where by reference to a contract term…a person seeks to restrict liability to a specified sum of money, and the question arises (under this or under any other Act) whether the term or notice satisfies the requirement of reasonableness, regard shall be had in particular…to (a) the resources which he could expect to have available to him for the purpose of meeting the liability should it arise; and (b) how far it was open to him to cover himself by insurance.

Standard form contracts do therefore have their drawbacks. Unfortunately the Act gives no guidance about when a contract, which is used several times, becomes "standard form". It is, however, at least arguable that standard terms offered by one party to another for purchases over the Internet are standard form [5] .

5. One further piece of legislation which should be noted is the Unfair Terms in Consumer Contracts Regulations 1994 and 1999, which implement EC Directive (93/13/EEC) on Unfair Terms in Consumer Contracts. This relates to terms in general and not just to exclusion clauses; but it applies only to contracts between consumers and sellers or suppliers who deal on terms which have not been individually negotiated. It is therefore relevant to Internet shopping by individuals, but not to the acquisition of bespoke software by a business and thus is not dealt with in this chapter. See further Koffman and MacDonald: *The law of contract,* 3rd edn (Tolly Publishing, 1998), Chapter 11.

There are a number of cases on the Unfair Contract Terms Act and on the requirement of reasonableness. The cases must, however, be treated with caution for the decision in each turns on its facts and just because one case is decided one way it does not mean that all others will follow. One example is *St Albans City and District Council v International Computers Ltd* [1996] 4 All ER 481. The facts were that the council had ordered a computer system from ICL to enable them to compute the Community Charge for the forthcoming year. ICL insisted on using its standard terms and conditions which stated that its liability "will not exceed the price or charge payable for the item of Equipment, Program or Service in respect of which liability arises or £100,000 (whichever is the lesser)…". Errors in the software and incorrect advice from ICL's project manager resulted in an overestimate of the population of the area, an undercharge of residents and a loss to the council of £1.3 million.

The judge at the initial hearing, Scott Baker J. found that the software was not fit for the purpose for which it was provided and that ICL's project manager had been negligent. ICL were therefore in breach of contract. As ICL had dealt on the basis of their standard terms and conditions, Section 3 of the Unfair Contract Terms Act 1977 applied and the clause limiting liability had to be measured against the requirement of reasonableness. Applying the factors listed in Section 11(4) and Schedule 2 (above), the judge noted that ICL were a substantial organization with world wide product liability insurance of £50 million; that all potential suppliers of the system dealt on similar standard terms; that the council was under pressure to install the system before the Community Charge was introduced; that although the council was a business and not a consumer, that it did not usually operate in the same commercial field as a normal business and that it would be

impractical for it to insure against commercial risks. On balance, the judge found that the clause limiting liability to £100,000 was not reasonable and was therefore ineffective. ICL appealed, but the Court of Appeal agreed with the decision of Mr Justice Scott Baker. However, as stated above, whatever the final outcome in this case, its value as a precedent will be limited, for each case turns on its facts. It cannot be concluded that henceforth no limitation of liability clause in a contract for the supply of a computer system will be valid. It does, however, show the pitfalls surrounding contract negotiation and a need to tailor such clauses to the circumstances. Dogged reliance on standard terms and conditions can prove expensive.

5.4 Further reading

There are a large number of books on computer contracts, most of which are written mainly for lawyers. The following is a list of books which are easier to read and understand, and yet give insight into the difficulties and practicalities.

Bainbridge, 1996. *Introduction to computer law ,* 4th edn. Chapters 14–22. London: Pitman.

Reed (ed.), 2000. *Computer law ,* 4th edn. Blackstone Press (especially the chapters on hardware contracts, software contracts and liability).

Rowland and Macdonald, 1997. *Information technology law* . Cavendish Publishing, Chapters 3 and 4.